



Waveforms and Spectrograms Activity Guide

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NEOM

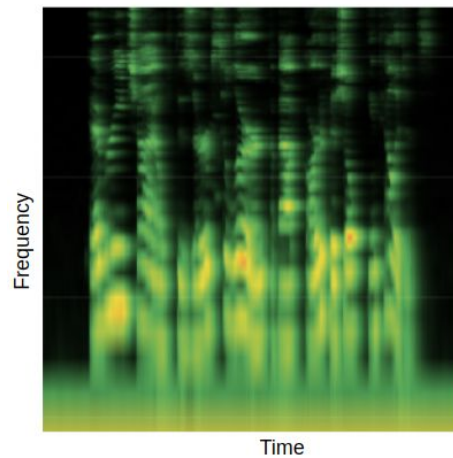
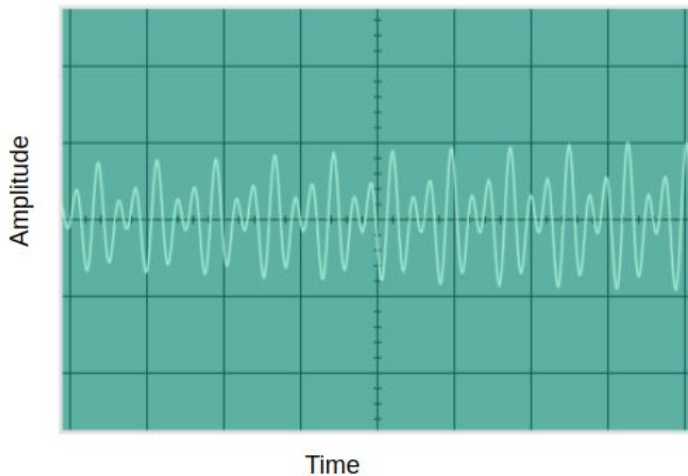


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Waveforms and Spectrograms Activity

Summary

Learn about the audio signal that is the input to the first stages of speech recognition by experimenting with waveforms and spectrograms. Use the laptop's microphone to capture musical tones and human speech.



Grade Bands: 3-5, 6-8, 9-12

Learning Objectives

Students will be able to:

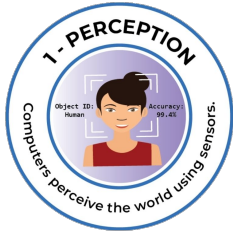
- Use software tools to demonstrate concepts such as amplitude, frequency, chords, and harmonics.
- Explain the difference between oscilloscope (waveform) and spectrogram displays.
- Demonstrate how different sounds in English result in different spectrogram patterns.

Materials/Resources Required

- Laptop with web browser and microphone
- Access to oscilloscope demo: <https://academo.org/demos/virtual-oscilloscope/>
- Access to spectrogram demo: <https://spectrogram.sciencemusic.org/>
- Optional: a musical instrument to play simple tones, or an online keyboard such as:
 - <https://www.onlinepianist.com/virtual-piano>
 - <https://virtualpiano.net/>

Guidelines and Standards

AI4K12 Guidelines



Big Idea 1: Perception

1-A-iii.6-8 Digital Encoding

1-B-ii.9-12 Feature Extraction

1-B-iii: Abstraction Pipeline: Language

Other Relevant Educational Standards

Next Generation Science Standards

- 1-PS4-1.** Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.
- 4-PS4-1.** Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.
- HS-PS4-1.** Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media.

Other Relevant Activities

1. SpeechDemo Activity
2. Machine Translation Activity

Vocabulary Terms

Flashcards for these vocabulary terms are located at the end of this guide.

- Amplitude
- Formant
- Frequency
- Harmonic
- Hertz or Hz
- Octave
- Oscilloscope
- Sine wave
- Spectrogram
- Waveform

Preparation

Setting Up

1. Make sure the computers can access the oscilloscope and spectrogram web pages, and give them access to the microphone.
2. Make sure the computers can access one of the digital keyboard web pages if you decide to use a digital keyboard to play notes.
3. If you're going to use an actual musical instrument to play notes, select an instrument that produces a simple waveform without a lot of overtones. The best choice is an electronic keyboard, which can produce a pure sine wave. Next best would be a piano (turn off the sustain feature if you can). Less good choices would be a trumpet or guitar because their waveforms are more complex.
4. An instrument with a real keyboard is better because most browser-based virtual keyboards don't provide a way to play chords, but you may be able to fake it by playing two notes in rapid succession and using sustain.

Introduction to Waveforms: What to Tell Your Students

Sound is a rapid variation in air or water pressure. There is no sound in outer space because there is no air, but there is sound underwater.

A microphone converts pressure variations into voltage variations that a computer can detect.

The computer measures the voltage thousands of times each second.

If we plot the voltage values over time we can see the “waveform”: the pattern of pressure changes that we perceive as sound.

An oscilloscope is a device for visualizing waveforms. Rapid changes back and forth (i.e., in pressure or voltage) are called “oscillations”, so an oscilloscope is a device for seeing oscillations.

“Frequency” refers to how quickly the air pressure is oscillating. Frequency determines the pitch of the sound. Fast oscillations (high frequencies) result in higher pitch sounds.

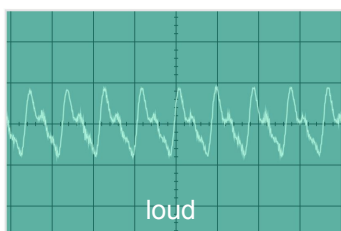
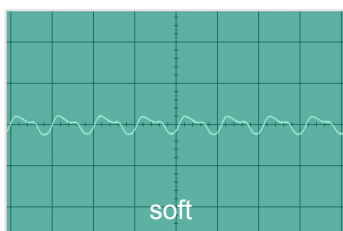
Examining Waveforms (Oscilloscope Demo)

1. Open <https://academo.org/demos/virtual-oscilloscope/>
2. Change the “Seconds/div” setting from 1 ms to 5 ms; see the red box (b) in the figure at right.
3. Speak into the microphone to see the waveform of your voice.
4. **Students may ask** Why did we need to set the time scale to 5 milliseconds?

Answer: we’re working with relatively low frequency sounds. If we use a faster time scale we won’t see much variation in the signal because it’s not changing quickly enough. With 5 milliseconds per division, the display has room for 40 milliseconds of signal, which is enough to clearly display a low frequency tone. But it’s not enough time to fit an entire word.

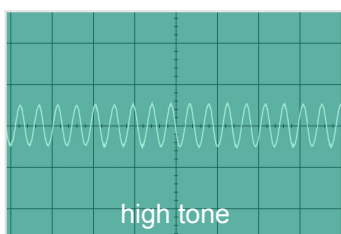
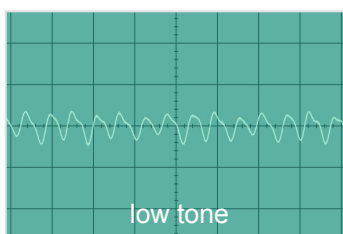
Try This!

- 👉 5. **Explore amplitude variation:** Try speaking more softly and then more loudly. How does the shape of the waveform change? (Answer: louder sounds produce a larger signal: you will see higher peaks and lower troughs in the waveform.)

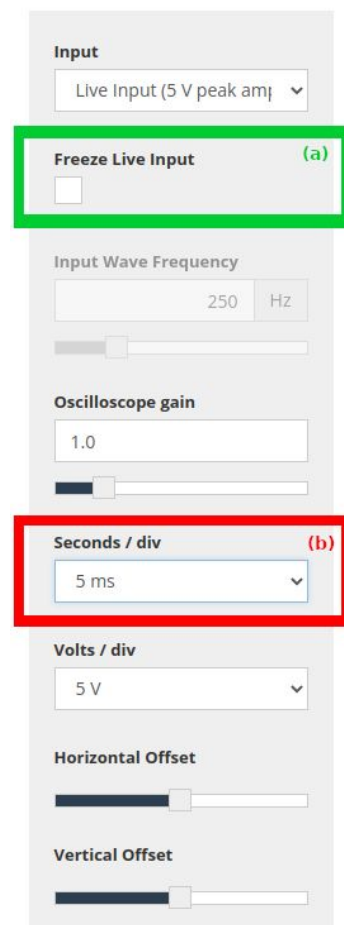


Try This!

- 👉 6. **Explore frequency variation:** Try singing or whistling a steady low-pitch tone and then a steady high-pitch tone.



Ask your students: How does the shape of the waveform change with the pitch (frequency)?

A screenshot of the Virtual Oscilloscope interface. The "Input" dropdown is set to "Live Input (5 V peak am)". The "Freeze Live Input" checkbox is checked and labeled (a). The "Input Wave Frequency" is set to 250 Hz. The "Oscilloscope gain" is set to 1.0. The "Seconds / div" dropdown is set to 5 ms and is labeled (b). The "Volts / div" dropdown is set to 5 V. The "Horizontal Offset" and "Vertical Offset" sliders are at the zero position.

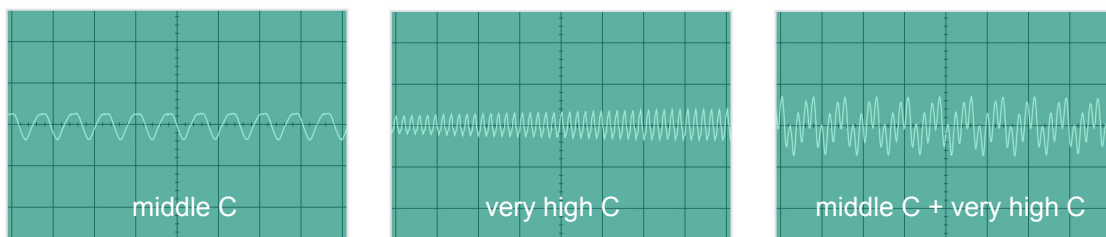
Answer: low frequency means slower oscillations, which appear as wider peaks and troughs, so fewer of them fit in the 40 millisecond wide display. With high tones, the pressure oscillates more rapidly, so the peaks and troughs are closer together, and more of them fit in the span of time shown on the display. Each vertical division line in the display is 5 milliseconds, or 1/200th of a second. So if a tone has two peaks between each vertical division, that would be a frequency of 400 cycles per second, or 400 Hertz.

7. **Students may ask:** When I sing or whistle a steady tone, the waveform shifts around and I sometimes see multiple lines crossing each other. What's going on?

Answer: This is an artifact of the way the oscilloscope display is updated. If you click on the "Freeze Live Input" checkbox (green box (a) in earlier figure) you will see that the actual waveform for a steady tone looks pretty much like a sine wave; there are no crossing lines.

Try This!

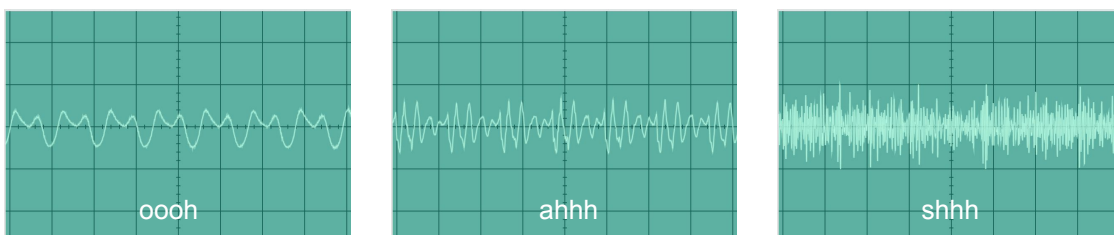
8. **Visualizing chords:** A single pure note like middle C looks like a sine wave; see the leftmost figure below. A higher note, such as the C two octaves above middle C, looks like a sine wave with a higher frequency; see the center figure below. When we play multiple notes at the same time, called a "chord", the waveforms add and produce a more complex shape, as in the rightmost figure below.



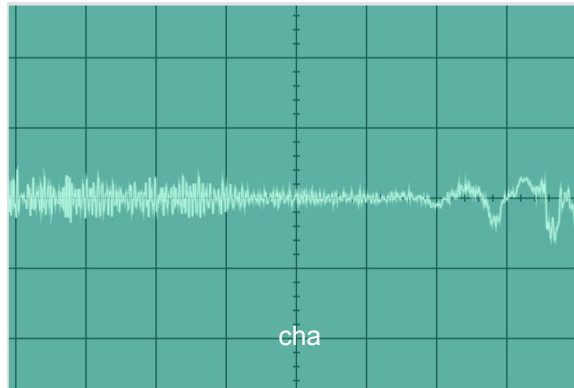
Using a keyboard (set the instrument type to "organ") to play chords, compare the waveforms for (1) middle C, (2) the C two octaves above that, and (3) the two notes played together. The chord is shown in the rightmost figure above. It looks like the middle C sine wave but with high frequency ripples superimposed. These ripples come from the very high C note.

Try This!

9. **Visualizing speech sounds:** Speech sounds produced by the vocal tract are much more complex than the pure tones produced by an electronic instrument. Try "oooh" (as in "tube") vs. "ahhh" (as in "father") vs. "shhh" (as in "shush") and notice the difference in the waveforms. The "oooh" and "ahhh" are low-frequency sounds relative to the hissing sound of the "shhh". A hiss is a mixture of high frequencies with no low frequencies.



10. **Visualizing syllables:** Below is the waveform for “cha” (as in “chop”). You can see that it has a complex shape that changes over time. The “ch” sound on the left looks like a hiss and is a mixture of high frequencies, while the “ahh” vowel sound on the right is primarily a low frequency tone with little bits of higher frequencies mixed in.

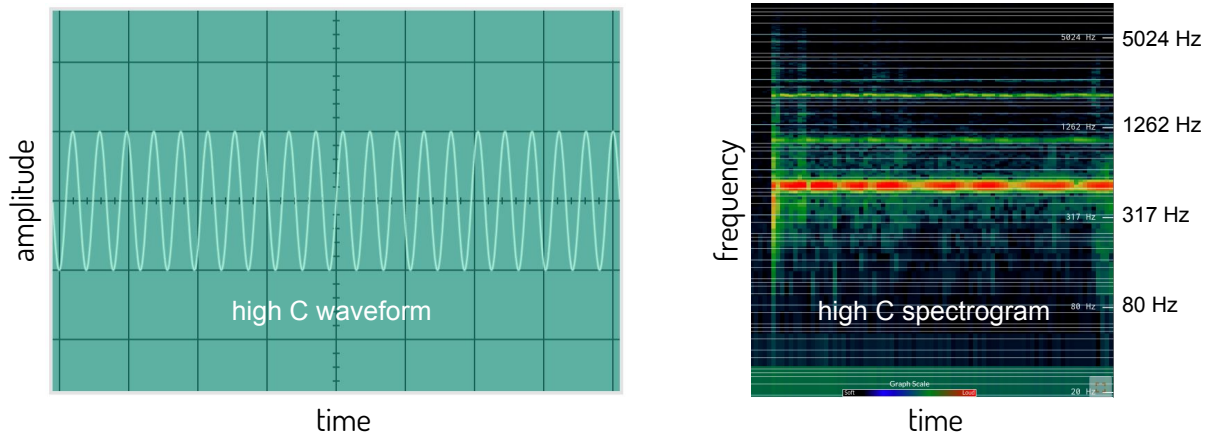


11. **Summary:** These experiments give some insight into the nature of the audio signal that is the input to the speech recognition process. We’ve seen that the waveform amplitude varies with loudness and the frequency varies with pitch (low note or high note). We’ve also seen that pure tones look like sine waves but richer sounds such as human speech appear as complex waveforms with multiple frequency components that are difficult to distinguish. In the next section we’ll explore a better way to represent and visualize these components to make it easier to tell sounds apart. This transformation of the waveform into a different representation is the first step in the speech recognition process.

Introduction to Spectrograms: What to Tell Your Students

We saw in the previous section that pure tones are sine waves, but when we mix several tones together, the waveform has a complex shape. It's hard to identify the different frequency components that contribute to the mixture just by looking at the waveform.

A spectrogram is another way to look at the audio signal. As with the oscilloscope display, the x-axis denotes time. But in a spectrogram, the y-axis indicates pitch (frequency) rather than amplitude. Since the speech signal can have contributions from multiple frequencies, instead of plotting a single line like a waveform, we use color to show the amount of energy in each frequency band. High energy, shown as yellow or red in the display, means that this frequency contributes significantly to the mixture. Here is the waveform for high C (one octave above middle C) along with the corresponding spectrogram:

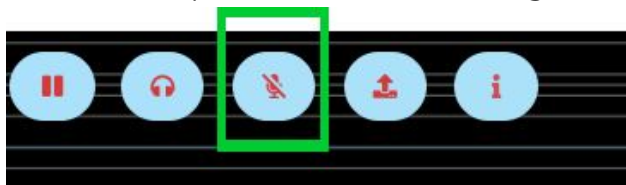


In the spectrogram display, a pure tone shows up as a thick red line. Low frequencies produce lines near the bottom of the display while high frequencies produce lines near the top. If we were to play a chord we would see multiple lines, one for each note of the chord.

For speech, the waveform is very complex. The major frequency components of speech (the frequency bands with highest energy) are called “formants”. Typically there are three major formants, labeled F1, F2, and F3. They move up and down in frequency depending on the speech sounds being produced, but F1 is always the lowest frequency band and F3 the highest.

Examining Spectrograms

1. Open <https://spectrogram.sciencemusic.org/>.
2. Click on the microphone icon twice to begin recording.



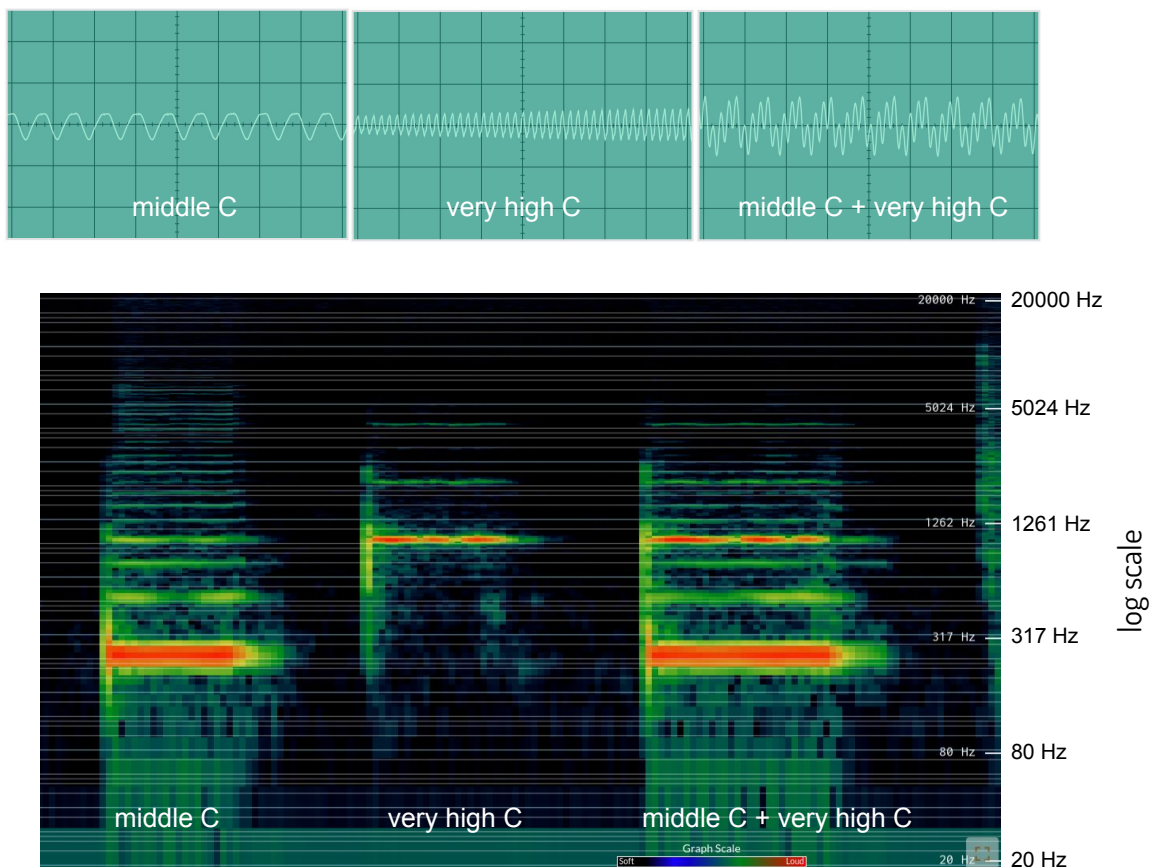
3. Speak into your laptop's microphone and observe the spectrogram.



4. Press the space key to freeze the display so you can examine the spectrogram more closely. Press the space key again to unfreeze it when you're ready to collect new audio.
5. Recall the waveforms we saw for middle C and very high C (two octaves above middle C). In the spectrogram, middle C shows up as a high energy band at around 256 Hertz, while very high C shows up as a high energy band at around 1024 Hertz. When we play both notes together, we will see both high energy bands.

Try This!

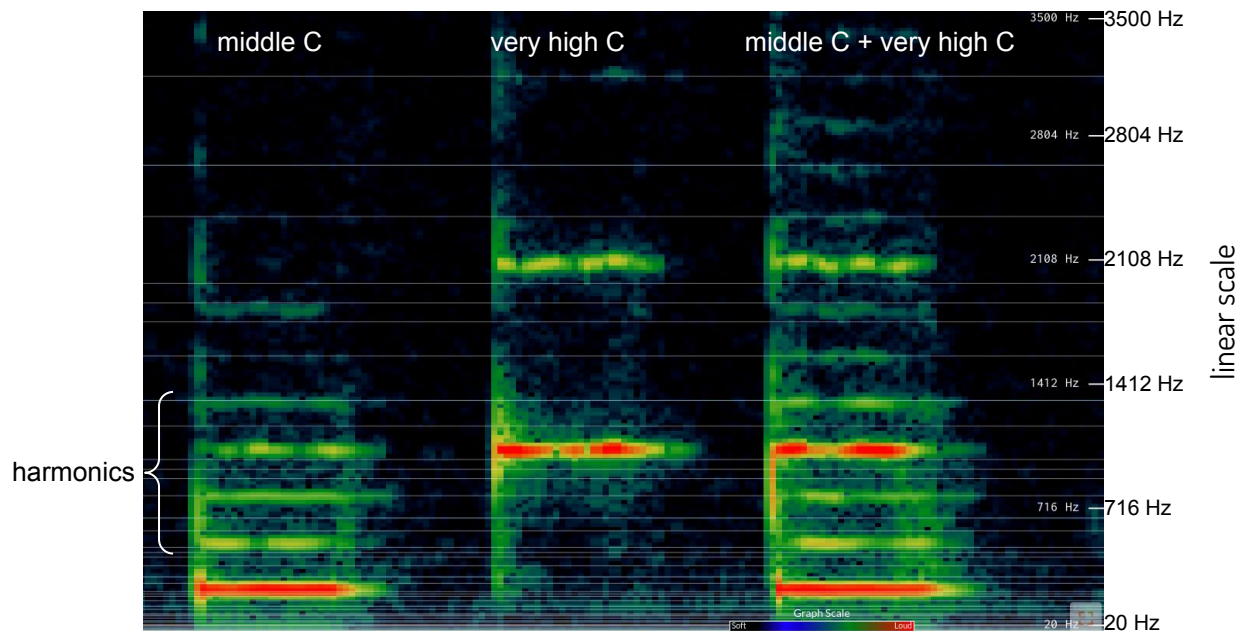
- 👉 6. Play middle C, then very high C, then the chord.



7. **Students may ask:** Why is the high energy band shown for middle C a much thicker red line than the high energy band for very high C?

Answer: This is because the spectrogram is plotted using a log scale for frequency. If we instead use a linear scale then the bands will be of equal thickness, as you can see in the next figure. Don't worry about how to switch the display to linear scale; we'll use the log scale for all our experiments because the log scale is better for displaying speech sounds.

Below is the same spectrogram as on the preceding page, but plotted using a linear frequency scale instead of a log scale. The division lines at right are evenly spaced every 696 Hertz. Notice that the high energy bands (red lines) for middle C on the left and very high C in the center are of equal thickness. The chord on the right combines the energy bands of both these notes.



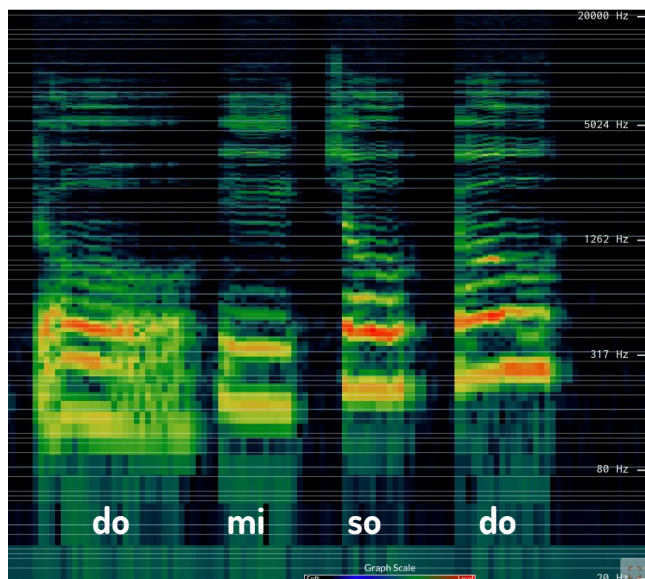
5. **Students may ask:** Why do we see above each high energy band some evenly-spaced (on the linear scale) weaker energy bands?

Answer: These are the “harmonics”. They appear at integer multiples of the fundamental frequency. For example, middle C at 256 Hertz has harmonics at $2x$, $3x$, and $4x$ that frequency, or 512, 768, and 1024 Hertz. Very high C at 1024 Herz has a harmonic at 2048 Hertz and a much weaker one at 3096 Hertz.

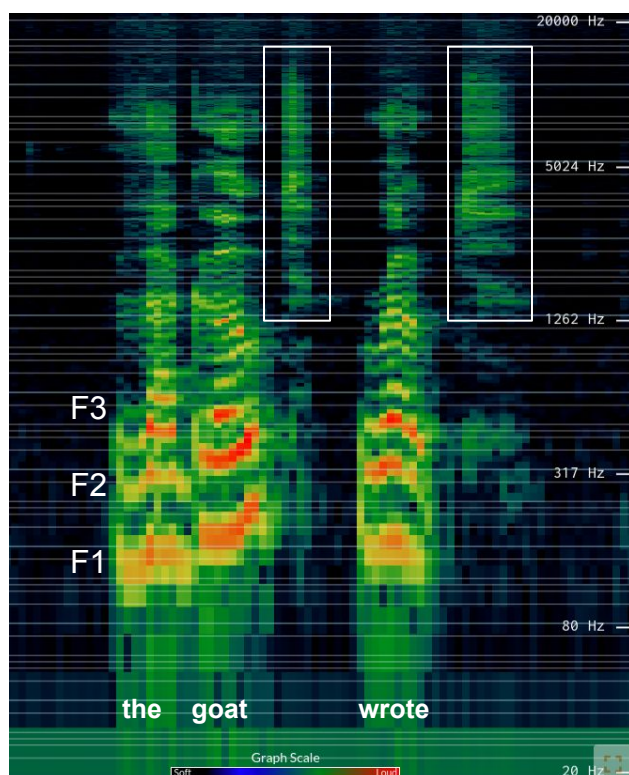
Harmonics are part of what distinguishes one type of instrument from another. We used an organ to produce these plots because its harmonics are simple. Reed instruments such as clarinets and saxophones have more complex harmonics, and the human vocal tract is even more complex.

Try This!

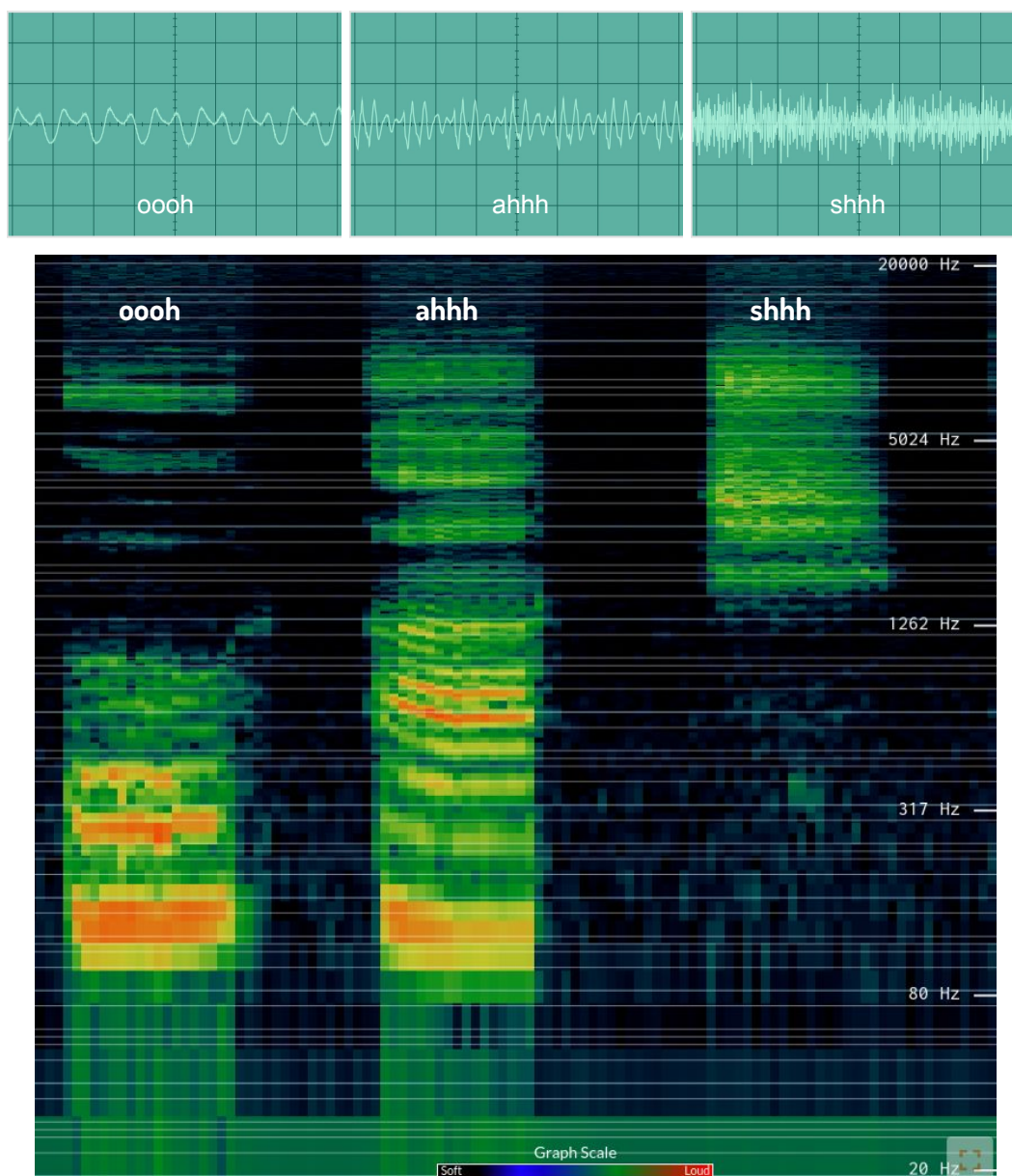
6. **Singing.** The spectrogram below shows a person singing a scale middle C-E-G-high C, or do-mi-so-do. Try singing some scales and see what that looks like.



7. **Formants.** Formants are the major frequency components (high energy bands) in the speech signal. They are labeled, F1, F2, F3, etc., with F1 being the lowest frequency. Formants move up and down with intonation (voice pitch). Early speech recognition systems paid a lot of attention to formants, using algorithms called “formant trackers”. Modern systems based on neural networks take a more holistic approach and look at the entire spectrogram. The spectrogram below shows the formants when a person is saying “the goat wrote”, with “goat” having a rising intonation and “wrote” a falling intonation. The white boxes enclosing hiss-like sounds (no formants visible) denote the puff of air released after the “t” in “goat” and “wrote”.



8. **Telling sounds apart.** If we look carefully at the spectrograms for different sounds we can notice features that distinguish one sound from another. Consider the sounds “oooh” (as in “tube”), “ahhh” (as in “father”), and “shhh” (as in “shush”):

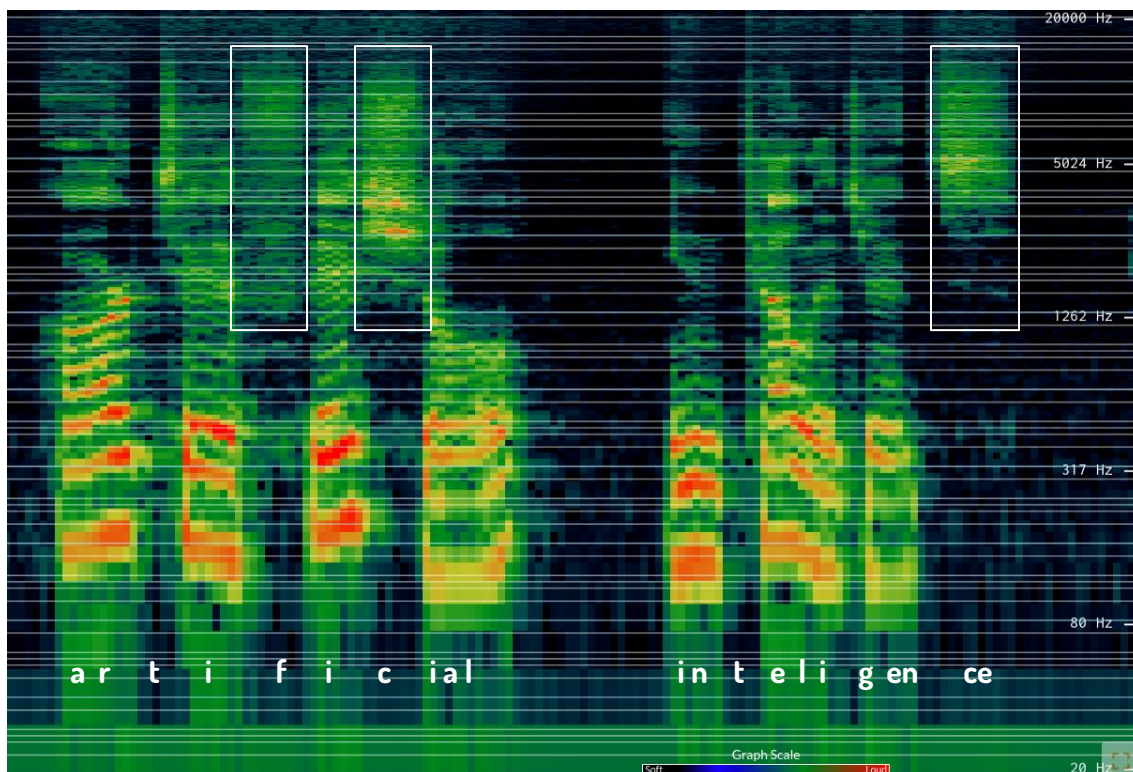


You can see that “oooh” is a low frequency sound, while “ahhh” has both low and high frequency components, and “shhh” is all high frequency hiss.

Try This!

9. Try making your own spectrograms of different vowel sounds and observing how they differ. Vowels are easier to distinguish than consonants because they can be held like a sung note, whereas consonants are brief events whose distinguishing features can be quite subtle. For example, the spectrograms for “pat” and “cat” or “do” and “goo” look nearly identical. This is one of the things that makes speech recognition difficult.

10. **Analyzing speech.** Taking everything we've learned about spectrograms we can look at a segment of human speech and tease apart some of the structure. Here is a spectrogram of someone slowly saying the phrase "artificial intelligence".



Because the speaker was pronouncing the words slowly it is easy to see the gaps between syllables, and also the longer gap between the two words. We can distinguish the spectrogram patterns for the "a" in "artificial" vs. the short "i" in both "artificial" and "intelligence". We can also point out the hissing patterns in the "f" and "sh" sounds in "artificial" and the "s" sound at the end of "intelligence", marked by white boxes.

Try This!

11.  Speak a phrase, record the spectrogram, and analyze the structure, pointing out word boundaries, the hissing patterns of sibilants (like "s") and fricatives (like "f"), and the different vowel sounds.,
12. **Summary:** the transformation from a waveform to a spectrogram representation is the first step in the speech recognition process. The next steps are to recognize individual sounds (vowels and consonants) and then recognize morphemes (word fragments such as "un-" or "-ing") and words. Early speech recognition systems worked this way. Modern systems, built using neural networks, take a more integrated approach instead of separating these processes into distinct steps.

Speech recognition is difficult because the speech signal is noisy and ambiguous. It can be difficult to tell consonants apart, and there are substantial variations from one speaker to another. Even the same speaker won't pronounce a word exactly the same way every time. Computers use several types of linguistic knowledge to help determine the most likely transcription of the speech signal. This knowledge includes vocabulary lists (to reject non-words) and grammar rules (to downvote hypotheses that are ungrammatical).

Assessments

To be determined.

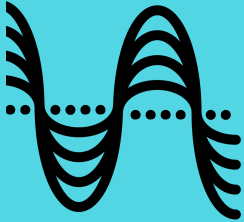
Other resources

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. This work was funded by a grant from NEOM Company.

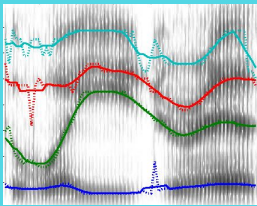
Visual design by Pam Amendola.

Vocabulary Flash Cards



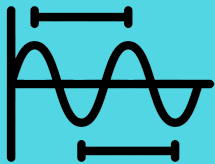
AMPLITUDE

The size of a signal. For sound, the amplitude is the loudness. For electrical signals it's the size of the voltage change.



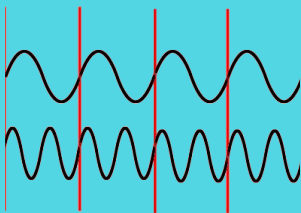
FORMANT

A frequency band in a speech spectrogram containing relatively high energy. Normal speech is composed of several formants.



FREQUENCY

The rate of variation in a periodic signal. For sound this corresponds to pitch. Frequency is measured in Hertz, or cycles per second.



OCTAVE

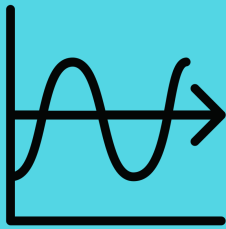
The distance between notes when one is twice or half the frequency of the other. A above middle C (440 Hz) and A below middle C (220 Hz) are one octave apart.

Vocabulary Flash Cards



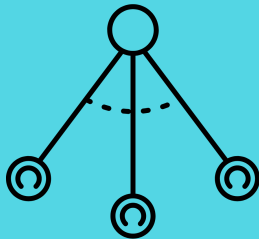
HARMONIC

An integer multiple of a frequency, e.g., A above middle C is 440 Hz, and its second and third harmonics are at 880 and 1320 Hz.



HERTZ or Hz

A unit of frequency corresponding to cycles per second. The note A above middle C is 440 Hertz, or 440 cycles per second.



OSCILLATION

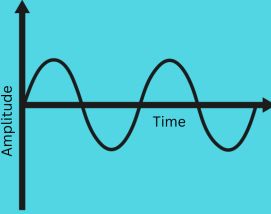
Rhythmic variation back and forth between two extremes. Sound is oscillations in air pressure.



OSCILLOSCOPE

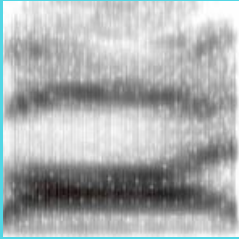
A device for visualizing oscillations in electrical signals. A microphone converts sound to voltage, allowing it to be seen with an oscilloscope.

Vocabulary Flash Cards



SINE WAVE

A smoothly varying waveform with constant amplitude and frequency. The shape of the wave follows the sine function.



SPECTROGRAM

A visualization of the energy in a signal at various frequencies, over time. Converting a waveform to a spectrogram is the first step in speech recognition.



WAVEFORM

A signal represented as an amplitude over time. Sine waves are waveforms corresponding to simple tones, while speech waveforms are more complex.